

Problem Set for 20 March 2025

In LADR4e,

1. $\mathcal{L}(U, V) := \text{Hom}_{\mathbb{F}}(U, V)$, and $\mathcal{L}(U) := \text{End}_{\mathbb{F}}(U)$ as the ground field $\mathbb{F}$ is fixed;
2. $\text{null} = \ker$, and $\text{range} = \text{im}$;

Problems from LADR4e.

1. (3A-17) Suppose V is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$.
 - A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a two-sided ideal of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.

答: 使用矩阵会清楚一些. 记 S 是 $\mathbb{F}^{n \times n}$ 的非零子集, 若

$$\forall A, B \in \mathbb{F}^{n \times n}, \forall M \in S : AMB \in S,$$

则任取 M 的非零分量, 得某一 $E_{i,j} \in S$. 此时

$$E_{s,t} = E_{s,i}E_{i,j}E_{j,t} \in S,$$

因此只能有 $\mathbb{F}^{n \times n} = S$.

- **(optional)** Find a non-trivial two-sided ideal of $\mathcal{L}(\mathbb{F}[x])$.

答: 所有 **im** 维数有限的映射构成双边理想, 见 (3B-23) 证明. 这一理想非平凡, 因为其既非零 0 , 又非 $\mathcal{L}(\mathbb{F}[x])$ (如 $\text{id}_{\mathbb{F}[x]}$ 不在理想内).

2. (3B-22) Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

- **(optional)** Both U and V can be ∞ -dimensional. The inequality holds as long as $\dim \text{null } S + \dim \text{null } T < \infty$.

答: 定义 U 子空间到 V 子空间的线性映射

$$\varphi : \ker ST \rightarrow \ker S, \quad u \mapsto T(u).$$

由定义, $\ker \varphi$ 是 $\ker T$ 的子空间, 从而 $\dim \ker ST < \infty$. 特别地,

$$\dim \ker ST = \dim \underbrace{\ker \varphi}_{\ker T \text{ 的子空间}} + \dim \underbrace{\operatorname{im} \varphi}_{\ker S \text{ 的子空间}} \leq \dim \ker S + \dim \ker T.$$

3. (3B-23) Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{range} ST \leq \min\{\dim \operatorname{range} S, \dim \operatorname{range} T\}.$$

- **(optional)** Both U and V can be ∞ -dimensional. The inequality holds as long as $\dim \operatorname{range} S < \infty$ or $\dim \operatorname{range} T < \infty$.

答: 一方面, $\operatorname{im}(ST)$ 是 $\operatorname{im}(S)$ 的子空间; 另一方面, S 给出了 $\operatorname{im}(T)$ 到 $\operatorname{im}(ST)$ 的满射. 明所欲证.

4. (3D-17) Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}(T) = ST$ for $T \in \mathcal{L}(V)$.

1. Show that $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.
2. Show that $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$.
3. **(optional)** Write down the matrix form of $\mathcal{A}$ under the basis $\{v_{i \cdot n + j} := E_{i,j}\}_{1 \leq i,j \leq n}$.

答: 见高等代数解题方法(第2版) 许甫华 张贤科 编著, 第六章T28 与T30. 书籍在 [github](#) 上, 此处就不费笔墨了.

注: 一个简便的记号是 Kronecker 积, 约定

$$A_{m \times n} \otimes B_{s \times t} = \begin{pmatrix} a_{1,1}B & a_{1,2}B & \cdots & a_{1,n}B \\ a_{2,1}B & a_{2,2}B & \cdots & a_{2,n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}B & a_{m,2}B & \cdots & a_{m,n}B \end{pmatrix}_{ms \times nt},$$

那么 $(A \otimes B) \cdot (C \otimes D) = AC \otimes BD$.

Since there are some [solutions](#) on the website, it is recommended to finish all optional problems.